# Neeraj Balachandar

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2026 | 8.67 |
| Kuriakose Elias English Medium School | Class XII (Kerala State board) | 2022 | 98.00% |
| Marian Senior Secondary School | Class X (Central Board of Secondary Education) | 2020 | 96.00% |

## RELEVANT COURSES

- **Mathematics**: Matrix Theory, Linear Algebra, Probability and Stochastic Processes, Information Theory
- **Dynamics & Control**: Control Systems, Optimal Control, Convex Optimization, Spacecraft Dynamics and Control
- **Computational Mechanics**: Incompressible Fluid Flows, CFD, FEM, Scientific Parallel Computing
- **Robotics**: Autonomous Robotics, Soft Robotics
- **Learning & Decision Making**: Machine Learning, Reinforcement Learning, Multi-Agent Systems, Game Theory

## SKILLS

- **Programming Languages**: Python, Julia, MATLAB, C, Embedded C
- **Tools**: ROS, Gazebo, MuJoCo, Paraview, Simulink, SolidWorks, Ansys Fluent, Simulia Abaqus, LaTeX, Github, OpenSim, Linux OS, VS Code, QP solvers

## PROJECTS

A Geometry-Informed, Multi-Task Neural Operator for Nonlocal Kernel Interactions in Lagrangian Flows: GitHub, Dr. Vishnu R. Unni

- Formulated and implemented an **operator learning** framework, modifying the existing Geometry-Informed Neural Operator to perform **multi-task prediction** in a **shared Fourier latent space**.
- The architecture was validated by learning two coupled tasks from a common latent space: **vortex-particle evolution** in the **Lagrangian** frame and **Eulerian velocity-field reconstruction**, exploiting their shared underlying **Poisson PDE** dynamics, through an operator-theoretic formulation.
- The architecture is compared against the classical **Fast Multipole Method** as a data-driven surrogate for speedup while maintaining accuracy. Systematically studied **input-output representations** through parameter sweeps.

An Aeroelastic Solver Integrating Reformulated-Vortex-Particle and Finite-Element Methods across Non-Conforming Interfaces: GitHub, Dr. Vishnu R. Unni, Dr. Niranjan Ghaisas

- We developed the framework **VarFlExI**, a variable-fidelity **partitioned solver** that uses a **meshless** scheme to solve the Navier-Stokes equations with a reformulated **vortex particle method**, coupled with the **FEniCS** to solve dynamic aeroelasticity problems using **Generalized-**.
- The solver features strong **work-conservation** techniques that exploit the **Common Refinement Method** (CRM) or **Wendland-kernel**-based interpolation (RBF) to transfer aerodynamic forces and structural geometry across **non-conforming** and **multi-representative discretizations** through a common interface.
- Designed the framework primarily for **vortical-structure-dominated** flows, with variable-fidelity configurations that enable substantially **lower computational** cost than fully meshed solvers, reducing simulation times from days to hours while retaining essential **bi-directionally coupled** aeroelastic dynamics.

Nonlinear Model Predictive Control via Sequential Convex Programming for Drone-to-Drone Docking: GitHub, Dr. Vishnu R. Unni

- Developed a **receding-horizon nonlinear model predictive control** framework for autonomous drone-to-drone docking, formulating docking as a finite-horizon constrained optimal-control problem and solving it through **Sequential Convex Programming** (SCP).
- Formulated a reduced-order nonlinear multirotor model augmented with wind-disturbance states, and iteratively

linearized the dynamics within a **trust-region SCP** framework with **virtual control** to obtain tractable **convex subproblems**.
- Integrated disturbance-aware target-state estimation and prediction using a **Kalman filter** with receding-horizon replanning, enabling trajectory generation under **noisy measurements** and **target drift** for both stationary and constant-velocity target motion, simulated in **MuJoCo**.
- Developed a unified set of kinematic, geometric, and **safety constraints**, including docking-cone capture geometry, chaser-target collision avoidance, and actuation constraints, with **nonconvex constraints** convexified into **second-order cone programs**.

Control-Aware Model Reduction and Morphology-Control Co-Design for Soft Dynamical Systems: (thesis), Dr. Vishnu R. Unni, Dr. Prakhar Gupta

- Developing a **soft-robot** simulation and action-state **RL-Gymnasium** for a generalized class of soft robots to learn **locomotion**.
- Investigating **morphology-dependent controllability**, relating changes in the robot's dynamics and actuation to its achievable locomotion behaviors.
- Developing a **reduced-order** representation of body-wake dynamics that preserves locomotion and **control-relevant** behavior.
- Extending the framework toward **morphology-control co-design**, where learned locomotion performance guides the evolution of the robot's body and control strategy.

Modeling and Control of Aerial Robots from Constrained Dynamics: GitHub 1/2, Dr. Vishnu R. Unni, Dr. Nithyanandan Kanagaraj

- Developed a mechanically **constrained multi-DOF** aerial robot **platform** enabling **safe**, **repeatable excitation** of selected degrees of freedom, with direct measurement of generalized coordinates and estimation of the net wrench generated by the vehicle. Derivation of the framework using **Euler-Lagrange** equations.
- Characterized the **observability** and **identifiability** limits induced by the kinematic constraints through **rank, null-space, and condition-number analysis**, and developed a framework for net-wrench reconstruction from constrained-motion measurements.
- Developing free-flight aerial robot dynamics models from constrained experimental data using **inverse-dynamics-based wrench estimation** and **Koopman** models, with evaluation under model-parameter and sensor uncertainty.

## EXPERIENCE

Swarm Rescue Challenge Intern, CIEDS-Institut Polytechnique de Paris: Supervised by Dr. David Filliat

- Developed a **Python**-based **multi-UAV** exploration and rescue framework for **unknown environments** under obstacle, communication, and **resource constraints**.
- Designed **decentralized** swarm navigation using **Artificial Potential Fields** for exploration and **Rapidly-exploring Random Trees** (RRT) for return-to-base transport, with nearby drones maintaining a coordinated polygon formation around a designated leader.
- Evaluated swarm performance across 4 simulated environments and compared clustering-based exploration strategies using DBSCAN and K-means; achieved an average of **4.5 rescues and 1.5 collisions per 2000 simulation steps**.
- Presented the system and challenge results in person at **Institut Polytechnique de Paris**; **ranked second** among Indian teams based on combined technical performance and presentation.

Undergraduate Researcher, **Dynamics and Control** Lab (**DysCo**), IITH: Supervised by Dr. Vishnu R. Unni

- Conduct research on the **modeling and control of complex dynamical systems**, spanning fluid-structure interaction, aerial robotics, and soft robotics.
- Develop **data-driven** models and **reduced-order** representations for prediction, system identification, and control of nonlinear dynamical systems.
- Design **optimization** and learning-based **safe control** methods for constrained dynamical systems.
- Develop **reproducible** scientific-computing **frameworks** in **Julia** and **Python** for simulations, and conduct experimental/hardware-in-loop validation for robotic systems.
- **Teaching Assistant** for **Spacecraft Dynamics and Control** (Spring 2026 - Dr. Vishnu R. Unni) and **Control Systems** (Fall 2026 - Dr. Safvan Palathingal); prepared assessment and assignment problems and evaluated student solutions.

Research Intern, **Indian Institute of Science**: Supervised by Dr. SN Omkar

- Investigated human neuromuscular response under **near-microgravity** loading using experimental **EMG** time-series and **motion-capture** data, with relevance to wearable and **assistive robotics**. Used **OpenSim** to validate musculoskeletal simulations under altered loading conditions.
- Designed and developed a **wearable exoskeleton** to reproduce near-microgravity loading conditions.

## SCHOLASTIC ACHIEVEMENTS

- Awarded the **Undergraduate Research Excellence Award**, given annually to one student in the Dept of Mechanical and Aerospace Engineering at IIT Hyderabad for outstanding research contributions, **2026**.
- Ranked **second** among Indian teams in the **Swarm Rescue Challenge 2025**, organized by **CIEDS-Institut Polytechnique de Paris**.
- Awarded the **Summer Research Fellowship** by the **Indian Academy of Sciences** for a research internship at the **Indian Institute of Science** in Summer **2024**.

## EXTRACURRICULAR

- Part of the University Basketball team (**Inter IIT** 2023-2025).
- Completed **Trinity Grade 2 Piano** level.